



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		IV / IV - B.Tech. I - Semester							
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CT4101	Fundamentals of Deep Learning	PC	3	0	0	3	30	70	100
B23Hs4101	Human Resources & Project Management	Hs	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23CT4112	Prompt Engineering	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	-	30	--	30
B23CT4113	Evaluation of Industry Internship /Mini Project	PR	-	-	-	2	-	50	50
TOTAL			19	01	02	21	240	540	780

	Course Code	Course
# PE-IV	B23CT4102	Information Security
	B23AD4102	Devops
	B23CT4103	Data Science using Python
	B23CT4104	Mobile Computing
	B23CT4105	Cyber Security
	B23CT4106	12 week MOOC Swayam/ NPTEL course recommended by the BoS
# PE-V	B23CT4107	Data Engineering
	B23CT4108	AI & NN
	B23CT4109	Snow Flake Cloud Analysis
	B23CT4110	Advance Computing
	B23CD4107	Big Data Analysis
	B23CT4111	12 week MOOC Swayam/ NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE or EEE or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE or EEE or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4101	PC	3	--	--	3	30	70	3 Hrs.
FUNDAMENTALS OF DEEP LEARNING								
(Common to CSD & CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	To understand Deep Learning concepts, techniques, and their real-world applications.							
2.	To understand CNN algorithms and the way to evaluate performance of the CNN architectures.							
3.	To apply RNN and LSTM models for prediction and classification in real-world problems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand Deep learning concepts and Historical Trends							K2
2.	Explain the MP neuron and Perceptron learning for linearly separable problems							K2
3.	Illustrate the mathematical methods to train MLP, Forward and Backward Propagation							K3
4.	Implementing the basic of Convolution operations and their role in DP models.							K3
5.	Analyze and implement advanced RNN architectures for sequence learning tasks.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Historical Trends in Deep Learning, The Many Names and Changing Fortunes of Neural Networks, Increasing Dataset Sizes, Increasing Model Sizes, Increasing Accuracy, Complexity and Real-World Impact.							
UNIT-II (10 Hrs)	Neuron & Neural Network: The Brain And The Neuron: Hebb's Rule, McCulloch and PittsNeuronandItsLimitations,NeuralNetworks,ThePerceptron:TheLearningRate,BiasInput, ThePerceptron Learning Algorithm, Linear Separability, Linear Regression.							
UNIT-III (10 Hrs)	Multi-Layer Perceptron: Going Forward: Biases, Back-Propagation and Error, The Multi-Layer Perceptron Algorithm, Initializing the Weights, Activation Functions, Sequential and Batch Training, LocalMinima, Picking up momentum, Mini batches and Stochastic Gradient Descent, Amount of Training Data, Number of Hidden Layers,Whento Stop Learning.							
UNIT-IV (10 Hrs)	Convolution Networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely StrongPrior, Variants of Basic Convolution Functions, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features, The Neuroscientific Basis for Convolutional Networks, Convolutional.							
UNIT-V (10 Hrs)	Sequence Modeling- Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence							

	Architectures, Deep Recurrent Networks, Recursive Neural Networks, The Challenge of Long-Term Dependencies, Echo State Networks, Leaky Units and Other Strategies for Multiple Time Scales, LSTM and Other Gated RNNs, Explicit Memory.
Text Books:	
1.	Deep Learning, Ian Goodfellow, YoshuaBengio, and Aaron Courville, MIT Press.
Reference Books:	
1.	Fundamentals of Deep Learning, Nikhil Buduma, 1st Edition, O'Reilly
e-Resources	
1.	https://www.oreilly.com/library/view/fundamentals-of-deep/9781491925607/
2.	https://cs231n.github.io/optimization-2/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	HS	2	--	--	2	30	70	3 Hrs.
HUMAN RESOURCE & PROJECT MANAGEMENT								
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)								
Course Objectives:								
1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.							
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.							
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.							
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.							
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.							K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.							K2
3.	Explain project management concepts, project networks, and monitoring processes.							K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.							K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.							K2
SYLLABUS								
UNIT-I (10Hrs)	HRM: Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.							
UNIT-II (10 Hrs)	HRD: HR accounting – Models - Concept of Training and Development – Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.							
UNIT-III (10 Hrs)	Basics of Project Management: Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals – Monitoring project progress – Project appraisal and Project selection-80-20 rules.							

UNIT-IV (10 Hrs)	Project Selection and Appraisal: Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	Project Implementation and Review: Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: —Personnel and Human Resource Management-Text and Cases, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, —Projects, Planning, Analysis, Selection, Financing Implementation and Review, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: —Human Resource and Personnel Management, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4102	PE	3	--	--	3	30	70	3 Hrs.

INFORMATION SECURITY

(For CSIT)

Course Objectives: The main objectives of the course is to provide students with:

1.	To understand the fundamentals of cloud computing, IoT, wireless networks, and mobile device security.
2.	To learn cloud and IoT security concepts, approaches, and protection mechanisms for secure communication and data management.
3.	To analyze security threats and vulnerabilities in wireless and mobile environments and apply suitable security measures.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Apply authentication, access control, and identity management techniques such as DAC, RBAC, and ABAC to secure computer systems and networks.	K3
2.	Analyze database and software security threats and apply suitable countermeasures to protect systems and data.	K4
3.	Apply secure coding practices and input validation techniques to prevent overflow attacks and software security vulnerabilities.	K3
4.	Analyze network security threats and apply intrusion detection, firewall, and public-key infrastructure techniques to secure network systems.	K4
5.	Apply security concepts and protection mechanisms for wireless networks, cloud computing, IoT environments, and mobile devices.	K3

SYLLABUS

UNIT-I (10Hrs)	User authentication and Access Control: Digital User Authentication Principle, Password Based Authentication, Remote User Authentication, Security Issues for User Authentication Access Control Principles, Subjects, Objects, and Access Rights, Discretionary Access Control, Example: UNIX File Access Control, Role-Based Access Control, Attribute-Based Access Control, Identity, Credential, and Access Management, Trust Frameworks.
UNIT-II (10 Hrs)	Database and Data Center Security: The Need for Database Security, SQL Injection Attacks, Database Access Control. Software Security: Types of Malicious Software, Advanced Persistent Threat, Propagation — Infected Content - Viruses, Propagation — Vulnerability Exploit - Worms, Propagation — Social Engineering — SPAM E-Mail, Trojans, Payload — System Corruption, Payload — Attack Agent — Zombie, Bots, Payload — Information Theft — Keyloggers, Phishing, Spyware, Payload — Stealthing — Backdoors, Rootkits, Countermeasures.

UNIT-III (10 Hrs)	Overflow Attacks - Stack Overflows, Defending Against Buffer Overflows, Other Forms of Overflow Attacks. Software Security Issues - Handling Program Input, Writing Safe Program Code, Handling Program Input.
UNIT-IV (10 Hrs)	Network Security: Denial-of-Service Attacks, Flooding Attacks, Distributed Denial-of-Service Attacks, Overview of Intrusion Detection, Honeypots, The Need for Firewalls, Firewall Characteristics and Access Policy, Types of Firewalls, Public-Key Infrastructure.
UNIT-V (10 Hrs)	Wireless, Cloud and IoT Security: Cloud Computing, Cloud Security Concepts, Cloud Security Approaches, The Internet of Things, IoT Security. Wireless Security Overview, Mobile Device Security.
Text Books:	
1.	Computer Security: Principles and Practice, W. Stallings, L. Brown, 4th edition, Pearson Education, 2018.
Reference Books:	
1.	Security in Computing, Pfleeger C.P., Pfleeger S.L., Margulies J. 5th edition, Prentice Hall, 2015.
2.	Network security essentials Applications and Standards, Stallings W. , 6th edition, Pearson Education, 2018.
3.	Security Engineering: A guide to building Dependable Distributed Systems, Anderson R.J., 2nd edition, John Wiley & Sons, 2008.
e-Resources	
1.	nptel.ac.in/courses/106106199
2.	12 YouTube Channels to Learn Cybersecurity for FREE - Career Swami

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4102	PE	3	--	--	3	30	70	3 Hrs.
DevOps								
(Common to AIDS & CSIT)								
Course Objectives: This course is designed to:								
1.	To describe the agile relationship between development and IT operations							
2.	To understand the skill sets and high-functioning teams involved in DevOps and related methods to reach a continuous delivery capability							
3.	To implement automated system update and DevOps lifecycle							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Describe the DevOps lifecycle, Agile principles, and the architecture of CI/CD pipelines.							K2
2.	Execute source code management tasks using Git and perform code quality analysis with SonarQube							K3
3.	Construct automated build pipelines and manage Jenkins architecture for Continuous Integration.							K3
4.	Deploy containerized applications using Docker and automate testing using Selenium							K3
5.	Orchestrate application deployments using Ansible and Kubernetes resource management.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to DevOps: Introduction to SDLC, Agile Model. Introduction to Devops. DevOps Features, DevOps Architecture, DevOps Lifecycle, Understanding Workflow and principles, Introduction to DevOps tools, Build Automation, Delivery Automation, Understanding Code Quality, Automation of CI/ CD. Release management, Scrum, Kanban, delivery pipeline, bottlenecks, examples							
UNIT-II (8 Hrs)	Source Code Management (GIT): The need for source code control, The history of source code management, Roles and code, source code management system and migrations. What is Version Control and GIT, GIT Installation, GIT features, GIT workflow, working with remote repository, GIT commands, GIT branching, GIT staging and collaboration. Unit Testing - Code Coverage: Junit, nUnit & Code Coverage with Sonar Qube, SonarQube - Code Quality Analysis.							
UNIT-III (10 Hrs)	Build Automation - Continuous Integration (CI): Build Automation, What is CI Why CI is Required, CI tools, Introduction to Jenkins (With Architecture), jenkins workflow, Jenkins master slave architecture, Jenkins Pipelines, PIPELINE BASICS - Jenkins Master, Node, Agent, and Executor Freestyle Projects & Pipelines, Jenkins for Continuous Integration,							

	Create and Manage Builds, User Management in Jenkins, Schedule Builds, Launch Builds on Slave Nodes.
UNIT-IV (10 Hrs)	Continuous Delivery (CD): Importance of Continuous Delivery, Continuous Deployment CD Flow, Containerization with Docker: Introduction to Docker, Docker installation, Docker commands, Images & Containers, DockerFile, running containers, Working with containers and publish to Docker Hub.
	Testing Tools: Introduction to Selenium and its features, JavaScript testing
UNIT-V (8 Hrs)	Configuration Management - ANSIBLE: Introduction to Ansible, Ansible tasks, Roles, Jinja templating, Vaults, Deployments using Ansible. Containerization Using Kubernetes (Openshift): Introduction to Kubernetes Namespace & Resources, CI/CD - On OCP, BC, DC & ConfigMaps, Deploying Apps on Openshift Container Pods. Introduction to Puppet master and Chef.
Textbooks:	
1.	Agarwal, Gaurav., Modern DevOps Practices: Implement, secure, and manage applications on the public cloud by leveraging cutting-edge tools, 2nd Edition, Packt Publishing, 2024.
2.	Verma, Saurabh., Learning DevOps: Accelerate your organization's digital transformation by adopting DevOps practices, 2nd Edition, Packt Publishing, 2019.
Reference Books:	
1.	Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10
2.	Gene Kim Je Humble, Patrick Debois, John Willis. The DevOps Handbook, 1st Edition, IT Revolution Press, 2016.
3.	Verona, Joakim Practical DevOps, 1st Edition, Packt Publishing, 2016.

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B23CT4103	PE	3	--	--	3	30	70	3 Hrs.
DATA SCIENCE USING PYTHON								
(Common to CSD & CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Knowledge and expertise to become a data scientist.							
2.	Essential concepts of statistics and machine learning that are vital for data science.							
3.	Suitability and limitations of tools and techniques related to data science process							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the basic concepts of Data Science.							K2
2.	Explain the Machine Learning Activities on Data Science							K2
3.	Describe databases and NoSQL technologies for handling big data.							K2
4.	Illustrate the tools and applications on data science.							K3
5.	Apply the Visualization and development tools for data.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Data science, benefits and uses, facets of data, data science process in brief, big data ecosystem and data science. Data Science process: Overview, defining goals and creating project charter, retrieving data, cleansing, integrating and transforming data, exploratory analysis, model building, presenting findings and building applications on top of them							
UNIT-II (10 Hrs)	Applications of machine learning in Data science, role of ML in DS, Python tools like sklearn, modeling process for feature engineering, model selection, validation and prediction, types of ML, semi-supervised learning Handling large data: problems and general techniques for handling large data, programming tips for dealing large data, case studies on DS projects for predicting malicious URLs, for building recommender systems							
UNIT-III (10 Hrs)	NoSQL movement for handling Big data: Distributing data storage and processing with Hadoop frame work, case study on risk assessment for loan sanctioning, ACID principle of relational databases, CAP theorem, base principle of NoSQL databases, types of NoSQL databases, case study on disease diagnosis and profiling.							
UNIT-IV (10 Hrs)	Tools and Applications of Data Science: Introducing Neo4jfor dealing with graph databases, graph query language Cypher, Applications graph databases, python libraries like nltk and SQLite for handling Text mining and analytics, case study on classifying Reddit posts.							

UNIT-V (10 Hrs)	Data Visualization and Prototype Application Development: Data Visualization options, Cross filter, the Java Script Map Reduce library, Creating an interactive dashboard with dc.js, Dashboard development tools. Applying the Data Science process for real-world problem-solving scenarios as a detailed case study.
Text Books:	
1.	Davy Cielen, Arno D.B. Meysman, and Mohamed Ali, “Introducing to Data Science using Python tools”, Manning Publications Co, Dream tech press, 2016.
2.	Prateek Gupta, “Data Science with Jupyter” BPB publishers, 2019 for basics .
Reference Books:	
1.	Joel Grus “Data Science from Scratch”, OReilly, 2019
2.	Doing Data Science: Straight Talk from the Frontline, 1st Edition, Cathy O’ Neil and Rachel Schutt, O’Reilly, 2013
e-Resources	
1.	https://swayam.gov.in/nd1_noc19_cs60/preview
2.	https://towardsdatascience.com/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4104	PE	3	--	--	3	30	70	3 Hrs.
MOBILE COMPUTING								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	To understand the fundamentals of mobile communication.							
2.	To understand the architecture of various Wireless Communication Networks.							
3.	To understand the significance of different layers in mobile system Course Contents.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understanding the strong grounding in the fundamentals of mobile Networks.							K2
2.	Apply in MAC, Network, and Transport Layer protocols of Wireless Network							K3
3.	Demonstrate the Comprehend, design, and develop a lightweight network stack.							K3
4.	Analyze the Mobile Network Layer system working.							K4
5.	Examine WAP architecture, WAP model, and wireless telephone applications.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Wireless Networks: Applications, History, Simplified Reference Model, Wireless transmission, Frequencies, Signals, Antennas, Signal propagation, Multiplexing, Modulation, Spread spectrum, Cellular Systems: Frequency Management and Channel Assignment, types of hand-off and their characteristics.							
UNIT-II (10 Hrs)	MAC – Motivation, SDMA, FDMA, TDMA, CDMA, Telecommunication Systems, GSM: Architecture Location tracking and call setup, Mobility management, Handover, Security, GSM, SMS, International roaming for GSM, call recording functions, subscriber and service data management, DECT, TETRA, UMTS, IMT-2000.							
UNIT-III (10 Hrs)	Wireless LAN: Infrared vs. Radio transmission, Infrastructure, Adhoc Network, IEEE 802.11WLAN Standards, Architecture, Services, HIPERLAN, Bluetooth Architecture & protocols.							
UNIT-IV (10 Hrs)	Mobile Network Layer: Mobile IP, Dynamic Host Configuration Protocol, Mobile Transport Layer, Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/Fast recovery, Transmission/Time-out freezing, Selective retransmission, Transaction Oriented TCP.							

UNIT-V (10 Hrs)	Support for Mobility: Wireless Application Protocol: Architecture, Wireless Datagram Protocol, Wireless Transport Layer Security, Wireless Transaction Protocol, Wireless Session Protocol, Wireless Application Environment, Wireless Markup Language, WML Scripts, Wireless Telephone Application.
Textbooks:	
1.	Jochen Schiller, “Mobile Communication”, Second Edition, Pearson Education, 2008
2.	Cloud Computing, Theory and Practice, Dan C Marinescu, MK Elsevier.
Reference Books:	
1.	William Stallings, “Wireless Communications and Networks”, Second Edition, Pearson Education, 2004.
2.	C. Siva Ram Murthy, B. S. Manoj, “Adhoc Wireless Networks: Architectures and Protocols”, Second Edition, Pearson Education, 2008.
e-Resources	
1.	https://nptel.ac.in/courses/106/106/106106147/#
2.	https://www.tutorialspoint.com/android/index.htm



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B23CT4105	PE	3	--	--	3	30	70	3 Hrs.
CYBER SECURITY								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Identify security risks and threats in cyber space and take preventive steps.							
2.	Understand the tools and methods used in various cybercrimes.							
3.	Understand the digital forensics fundamentals including evidence collection and preservation.							
4.	Familiarize with computer forensics investigation tools and techniques.							
5.	Understand the legal perspectives and IT Act provisions related to cybercrime.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain the fundamentals of WAP architecture, model, and wireless applications.							K2
2.	Describe cybercrime tools and methods such as phishing, malware, DoS, SQL.							K2
3.	Apply cybercrime investigation tools for digital evidence password cracking.							K3
4.	Illustrate computer forensics fundamentals, tools, biometric forensics, and OS forensics.							K3
5.	Analyze legal aspects of cybercrime, including the IT Act, digital signatures.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Cybercrime: Definition and Origins, Cybercrime and Information Security, Who are Cybercriminals, Classifications of Cybercrime. Types of Cyber Threats: Cyberstalking, Cybercafe and Cybercrimes, Botnets, Attack Vectors. Mobile Threats: Proliferation of Mobile and Wireless Devices, Security Challenges Posed by Mobile Devices, Attacks on Mobile/Cell Phones, Network and Computer Attacks.							
UNIT-II (10 Hrs)	Tools and Attack Methods: Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography. Network Attacks: Sniffers, Spoofing, Session Hijacking, Buffer Overflow, DoS and DDoS Attacks, SQL Injection. Wireless Attacks: Attacks on Wireless Networks, Identity Theft (ID Theft), Foot printing and Social Engineering, Port Scanning, Enumeration.							
UNIT-III (10 Hrs)	Cybercrime Investigation: Introduction, Investigation Tools, eDiscovery, Digital Evidence Collection, Evidence Preservation. Digital Investigation Techniques: E-Mail Investigation, E-Mail Tracking, IP Tracking, E-Mail Recovery, Hands-on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers.							

	Recovering Deleted Evidence, Password Cracking.
UNIT-IV (10 Hrs)	Computer Forensics and Investigations: Understanding Computer Forensics, Preparing for Computer Investigations. Forensics Tools: Evaluating Computer Forensics Tools, Computer Forensics Software Tools, Computer Forensics Hardware Tools, Validating and Testing Forensics Software. Specialized Forensics: Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Graphics and Network Forensics, E-mail Investigations, Cell Phone and Mobile Device Forensics.
UNIT-V (10 Hrs)	Cybercrime Legal Perspectives: Introduction, Cybercrime and the Legal Landscape around the World, The Indian IT Act – Provisions and Amendments. Indian Cyber Law: Challenges to Indian Law and Cybercrime Scenario in India, Consequences of Not Addressing Weaknesses in IT Act, Digital Signatures and the Indian IT Act, Cybercrime and Punishment, Cyberlaw, Technology and Students – Indian Scenario.
Text Books:	
1.	Sunit Belapure & Nina Godbole, Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, WILEY, 2011.
2.	Nelson, Phillips and Enfinger Steuart, Computer Forensics and Investigations, Cengage Learning, New Delhi, 2009.
Reference Books:	
1.	Michael T. Simpson, Kent Backman and James E. Corley, Hands on Ethical Hacking and Network Defence, Cengage, 2019.
2.	John R. Vacca, Computer Forensics – Computer Crime Investigation, Firewall Media, New Delhi.
3.	Alfred Basta, Nadine Basta, Mary Brown & Ravinder Kumar, Cyber Security and Cyber Laws, Cengage, 2018.
e-Resources	
1.	https://www.coursera.org/learn/introduction-cybersecurity-cyber-attacks
2.	https://onlinecourses.swayam2.ac.in/e-learning/preview/nou19_cs08

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4107	PE	3	--	--	3	30	70	3 Hrs.
DATA ENGINEERING								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Understand the data engineering lifecycle from data generation to data serving.							
2.	Develop knowledge of data architectures, storage systems, and modern data engineering practices.							
3.	Equip students with skills in data ingestion, processing, and transformation for large-scale analytics.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Describe data engineering lifecycle.							K2
2.	Explain data architecture and data generation.							K2
3.	Explain data engineering storage abstractions.							K2
4.	Illustrate data ingestion process.							K2
5.	Discuss queries, modeling, transformation and serving.							K2
SYLLABUS								
UNIT-I (10Hrs)	Data Engineering Described: Data Engineering Lifecycle, Evolution, Data Engineering and Data Science; Skills & Activities- Data Maturity and the Data Engineer, The Background and Skills of a Data Engineer, Data Engineers Inside an Organization. The Data Engineering Life Cycle: The Data Lifecycle versus the Data Engineering Lifecycle, Storage, Ingestion, Transformation, Serving Data; Major Undercurrents across the Data Engineering Lifecycle.							
UNIT-II (10 Hrs)	Designing Good Data Architecture: Enterprise Architecture, Data Architecture, Principles of Good Data Architecture, Major Architecture Concepts and Types of Data Architecture, Who's Involved with Designing a Data Architecture. Data Generation in Source Systems: Sources of Data, Source Systems, Source System Practical Details- Databases, APIs, Data Sharing, Third-Party Data Sources, Message Queues and Event-Streaming Platforms.							
UNIT-III (10 Hrs)	Storage: Raw Ingredients of Data Storage, Data Storage Systems- Single Machine Versus Distributed Storage, Eventual Versus Strong Consistency, File Storage, Block Storage, Object Storage, Cache and Memory-Based Storage Systems, Streaming Storage, Data Engineering Storage Abstractions, Big Ideas and Trends in Storage- Data Catalog, Schema, Separation of Compute from Storage, Data Storage Lifecycle and Single-Tenant Versus Multitenant Storage.							

UNIT-IV (10 Hrs)	Ingestion: What is Data Ingestion, Key Engineering Considerations, Batch Ingestion Considerations, Message and Stream Ingestion Considerations, Ways to Ingest Data- Direct Database Connection, APIs, Message Queues and Event-Streaming Platforms, Managed Data Connectors, Moving Data with Object Storage, EDI, Databases and File Export, Practical Issues with Common File Formats, Web hooks, Web Interface, Web Scraping, Transfer Appliances for Data Migration.
UNIT-V (10 Hrs)	Queries, Modeling, and Transformation: Queries, Data Modeling, Transformations- Batch Transformations, Materialized Views, Federation, and Query Virtualization, Streaming Transformations and Processing, Serving Data for Analytics, Machine Learning: General Considerations for Serving Data, Analytics- Business, Operational, and Embedded; Machine Learning- What a Data Engineer Should Know About ML; Ways to Serve Data for Analytics and ML.
Text Books:	
1.	Fundamentals of Data Engineering: Plan and Build Robust Data Systems, Joe Reis and Matt Housley, O'Reilly.
Reference Books:	
1.	Data Engineering with Python: Work with massive datasets to design data models and automate data pipelines using Python , Paul Crickard, Packt Publishing
2.	Data Engineering Mining, Information and Intelligence, Edited by Yupo Chan, John R.Talbert, Terry M. Talley, ISBN 978-1-4419-0175-0 e-ISBN 978-1-4419-0176-7 DOI 10.1007/978-1-4419-0176-7 Springer New York Dordrecht Heidelberg London.
e-Resources	
1.	https://ieeexplore.ieee.org
2.	IEEE Xplore Digital Library
3.	https://freecomputerbooks.com/Fundamentals-of-Data-Engineering.html?utm_source=chatgpt.com

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4108	PE	3	--	--	3	30	70	3 Hrs.
ARTIFICIAL INTELLIGENCE AND NEURAL NETWORKS								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Introduce the foundations of Artificial Intelligence and Neural Networks.							
2.	Acquire knowledge on Soft Computing Concepts including Fuzzy Logic.							
3.	Learn various types of Neural Network architectures and training methods.							
4.	Gain knowledge to apply optimization strategies using Genetic Algorithms.							
5.	Design hybrid intelligent systems incorporating neural, fuzzy and genetic components.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the concepts of Artificial Intelligence, search algorithms.							K2
2.	Explain neural networks and suitable learning paradigms for real-world systems.							K2
3.	Interpret the Fuzzy Logic concepts including fuzzy sets, reasoning .							K3
4.	Apply Genetic Algorithms and Swarm Intelligence techniques for optimization.							K3
5.	Categorize neural networks, fuzzy systems, and genetic.							K4
SYLLABUS								
UNIT-I (10Hrs)	Soft Computing and Artificial Intelligence: Introduction to Soft Computing, Soft Computing vs. Hard Computing, Various Types of Soft Computing Techniques, Applications of Soft Computing. AI Search Algorithms: Problem Solving, State-Space Search, BFS, DFS, Heuristic Search – Best First Search, A* Algorithm, AO* Algorithm. Knowledge Representation: Predicate Calculus, Rules of Inference, Semantic Networks, Frames, Objects, Hybrid Models.							
UNIT-II (10 Hrs)	Artificial Neural Networks and Paradigms: Introduction to Biological and Artificial Neuron Models, Neural Network Architecture, McCulloch-Pitts Neuron. Learning Rules: Hebbian Learning, Perceptron Learning, Delta Rule. Network Models: Single Layer Perceptrons, Multilayer Perceptrons, Backpropagation Networks – algorithm, convergence and applications. Kohonen's Self-Organizing Networks, Hopfield Network, Associative Memory. Applications of Neural Networks.							
UNIT-III (10 Hrs)	Fuzzy Logic: Introduction, Crisp Sets vs. Fuzzy Sets, Membership Functions, and Basic Operations on Fuzzy Sets – Union, Intersection, and Complement. Fuzzy Relations, Composition, Fuzzy							

	If-Then Rules, Rule-Based Models and Linguistic Variables. Fuzzy Inference and Control: Mamdani and Sugeno Fuzzy Inference Systems, Fuzzification, Defuzzification Methods, Fuzzy Decision Making. Applications of Fuzzy Logic in Control Systems.
UNIT-IV (10 Hrs)	Genetic Algorithms: Introduction, Biological Inspiration, Genetic Algorithm Fundamentals – Encoding, Fitness Function, Selection Methods, Crossover Operators, Mutation. Evolutionary Programming, Genetic Programming – Parse Trees, Variants of GA. Swarm Intelligence: Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Artificial Bee Colony (ABC) Optimization. Applications of GA and Swarm Intelligence.
UNIT-V (10 Hrs)	Hybrid Intelligent Systems: Need for Hybrid Systems, Neuro-Fuzzy Hybrid Systems – Architecture and Working. Adaptive Neuro-Fuzzy Inference System (ANFIS) – Structure and Learning. Other Hybrid Approaches: Fuzzy Backpropagation Network, Genetic-Neuro Hybrid System, Genetic Algorithm based Backpropagation Network, Genetic-Fuzzy Hybrid Systems. Comparative Study and Application Domains.
Text Books:	
1.	Simon S. Haykin, Neural Networks and Learning Machines, Prentice Hall, 3rd Edition, 2009.
2.	S. Rajasekaran & G.A. Vijayalakshmi Pai, Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis & Applications, PHI, 2003.
Reference Books:	
1.	Jang J.S.R., Sun C.T. and Mizutani E., Neuro-Fuzzy and Soft Computing, Prentice Hall, 1998.
2.	Jacek M. Zurada, Introduction to Artificial Neural Systems, Jaico Publishing House, 1994.
3.	Timothy J. Ross, Fuzzy Logic with Engineering Applications, McGraw Hill, 3rd Edition, 2009.
e-Resources	
1.	https://nptel.ac.in/courses/117105084
2.	https://www.coursera.org/articles/artificial-neural-network

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4109	PE	3	--	--	3	30	70	3 Hrs.
SNOW FLAKE CLOUD ANALYSIS								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Understand cloud data warehousing concepts and the architecture of the Snowflake platform.							
2.	Apply Snowflake features for data loading, querying, and transformation using SQL.							
3.	Implement security, performance optimization, and monitoring tools in Snowflake environments.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain cloud data warehousing concepts and Snowflake architecture.							K2
2.	Apply Snowflake SQL for data loading, querying, and transformations.							K3
3.	Differentiate structured and semi-structured data processing in Snowflake.							K3
4.	Implement data security, sharing, and access control mechanisms.							K3
5.	Evaluate performance optimization and monitoring techniques.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Cloud Data Warehousing and Snowflake Introduction to cloud computing and data warehousing; traditional vs cloud data warehouses; overview of Snowflake platform; Snowflake architecture (storage, compute, cloud services); key features and advantages; virtual warehouses; databases, schemas, and tables; overview of Snowflake editions and use cases.							
UNIT-II (10 Hrs)	Snowflake Data Loading and SQL Operations Data loading concepts; stages (internal, external, named stages); file formats; COPY command; bulk data loading and unloading; Snowflake SQL basics; querying data; joins, aggregations, and functions; working with views and temporary tables; Data loading using Snowpipe (auto ingestion).							
UNIT-III (10 Hrs)	Semi-Structured Data and Data Transformation Introduction to semi-structured data (JSON, XML, Avro, Parquet); VARIANT data type; querying semi-structured data; flattening data; Data transformation techniques; streams and tasks for change data capture (CDC); time travel and data versioning; stored procedures and user-defined functions (UDFs); data pipelines and ETL/ELT processing; integration with external tools (BI & data tools).							
UNIT-IV (10 Hrs)	Data Sharing, Security, and Governance: Data sharing concepts; secure data sharing in Snowflake; roles and access control (RBAC); authentication and authorization; encryption and data protection; masking policies; Data governance and compliance features; data sharing with external systems and APIs; row-level and column-level security; data auditing and access monitoring.							

UNIT-V (10 Hrs)	Performance Optimization and Monitoring Query performance tuning; clustering and partitioning concepts; caching mechanisms; warehouse sizing and scaling; query profiling; monitoring usage and cost; resource monitors; best practices for optimization; cost optimization and billing management in Snowflake.
Text Books:	
1.	Snowflake: The Definitive Guide – Joyce Kay Avila
Reference Books:	
1.	Effective DevOps – Jennifer Davis (for DevOps integration concepts)
2.	Data Warehousing in the Age of Big Data – Krish Krishnan
3.	Cloud Data Management – John Rhoton
e-Resources	
1.	https://docs.snowflake.com
2.	https://quickstarts.snowflake.com
3.	https://youtu.be/JXd-Io443jU



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4110	PE	3	--	--	3	30	70	3 Hrs.

ADVANCE COMPUTING

(For CSIT)

Course Objectives: The main objectives of the course is to provide students with:

1.	Understand Grid Computing concepts including architecture, infrastructure, resource management,
2.	Explore Pervasive Computing environments, like device connectivity, human-machine interaction
3.	Gain foundational knowledge of Quantum Computing, qubits, superposition, , and cryptography.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand Grid Computing concepts, architecture, scheduling.	K2
2.	Describe and design Cluster Computing systems using appropriate architectures.	K2
3.	Apply Pervasive Computing concepts for device connectivity and HCI applications.	K3
4.	Interpret fundamental concepts of QC, including qubits, superposition.	K3
5.	Apply principles of Quantum Mechanics in the context of quantum computation.	K3

SYLLABUS

UNIT-I (10Hrs)	UNIT I: Grid Computing: Early Grid Activities, Current Grid Activities, An Overview of Grid Business Areas, Grid Applications Schedulers Resource Broker, Grid Infrastructure, Organizations Building and Using Grid-Based Solutions to Solve Computing, Data, and Network Requirements, Grid Architecture Grid Architecture and Relationship to Other Distributed Technologies, Autonomic Computing.
UNIT-II (10 Hrs)	UNIT II: Cluster Computing: Eras of Computing, A Cluster Computer and its Architecture, Clusters Classifications, Commodity Components for Clusters, Cluster Applications, Constructing Scalable Services Introduction, Environment, Resource Sharing, Flexible Load Sharing Algorithm.
UNIT-III (10 Hrs)	UNIT III: Pervasive Computing: Pervasive Computing concepts & Scenarios, Hardware & Software, Human - machine interface Device connectivity, Java for Pervasive devices, Application examples
UNIT-IV (10 Hrs)	UNIT IV: Quantum Computing Introduction: Quantum Information, Quantum Computers, State Description, Superposition, and Uncertainty, Measurements in Multiple Bases, Measurements of Superposition States, An Augmented Probabilistic Model—The

	Superposition Probability Rule, BB84—The Emergence of Quantum Cryptography, A Qubit of History.
UNIT-V (10 Hrs)	UNIT V: Quantum Mechanics: Vector Spaces, n-Dimensional Real Euclidean Vector Space, Linear Operators and Matrices, Hermitian Operators in a Complex n-Dimensional Euclidean Vector Space, n-Dimensional Hilbert Spaces—Dirac's Notation, The Inner Product in an n -Dimensional Hilbert Space, Tensor and Outer Products, Quantum States, Quantum Observables—Quantum Operators.
Text Books:	
1.	J. Joseph & C. Fellenstein, Grid Computing, PEA.
Reference Books:	
1.	Raj Kumar Buyya, High performance cluster computing, PEA.
2.	J.Burkhardt et .al, Pervasive computing, PEA.
3.	Vishal Sahni, Quantum computing, TMH.
4.	Marivesar, Approaching quantum computing, PEA.
e-Resources	
1.	nptel.ac.in/courses/106106232
2.	Lecture - 37 Pervasive & Ubiquitous Computing

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CD4107	PE	3	--	--	3	30	70	3 Hrs.
BIG DATA ANALYTICS								
(Common to CSD & CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Provide an overview of an exciting growing field of Big Data analytics.							
2.	Understand how to solve complex problems using Map Reduce							
3.	Introduce the tools required to analyze Big Data like Hadoop Map Reduce, Pig & Hive etc.,							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1.	Understand technologies and the need for file systems in Big Data analysis.							K2
2.	Explain the features of HDFS, Map Reduce to handle the Big Data.							K2
3.	Implement and analyze Map-Reduce programming model for better optimization.							K3
4.	Apply stream processing techniques for real-time data analysis in the Spark environment.							K3
5.	Interpret the need of Modern tools, viz., Pig and Hive, Hbase etc., on Big Data.							K3
SYLLABUS								
UNIT-I (10Hrs)	Data structures in Java: Linked List, Stacks, Queues, Sets, Maps; Generics: Generic classes and Type parameters, Implementing Generic Types, Generic Methods, Wrapper Classes, Concept of Serialization							
UNIT-II (10 Hrs)	Working with Big Data: Google File System, Hadoop Distributed File System (HDFS) Building blocks of Hadoop (Namenode, Datanode, Secondary Namenode, Job Tracker, Task Tracker), Introducing and Configuring Hadoop cluster (Local, Pseudo-distributed mode, Fully Distributed mode), Configuring XML files.							
UNIT-III (10 Hrs)	Writing Map Reduce Programs: A Weather Dataset, Understanding Hadoop API for Map Reduce Framework (Old and New), Basic programs of Hadoop Map Reduce: Driver code, Mapper code, Reducer code, Record Reader, Combiner, Practitioner							
UNIT-IV (10 Hrs)	Stream Memory and Spark: Introduction to Streams Concepts– Stream Data Model and Architecture , Stream computing, Sampling Data in a Stream , Filtering Streams ,Counting Distinct Elements in a Stream , Introduction to Spark Concept , Spark Architecture and components , Spark installation , Spark RDD(Resilient Distributed Dataset) – Spark RDD operations							
UNIT-V (10 Hrs)	Pig: Hadoop Programming Made EasierAdmiring the Pig Architecture, Going with the Pig Latin Application Flow, Working through the ABCs of Pig Latin, Evaluating Local and							

	Distributed Modes of Running Pig Scripts, Checking out the Pig Script Interfaces, Scripting with Pig Latin. Applying Structure to Hadoop Data with Hive: Saying Hello to Hive, Seeing How the Hive is Put Together, Getting Started with Apache Hive, Examining the Hive Clients, Working with Hive Data Types, Creating and Managing Databases and Tables, Seeing How the Hive Data Manipulation Language Works, Querying and Analysing data
Text Books:	
1.	Wiley & Big Java 4th Edition, Cay Horstmann, Wiley John Sons, INC
2.	Hadoop: The Definitive Guide by Tom White, 3rd.Edition, O'reilly
Reference Books:	
1.	Hadoop in Action by Chuck Lam, MANNING Publ.
2.	Hadoop for Dummies by Dirk deRoos, Paul C.Zikopoulos, Roman B.Melnyk,Bruce Brown, Rafael Coss
3.	Hadoop in Practice by Alex Holmes, MANNING Publ.
4.	Big Data Analytics by Dr. A.Krishna Mohan and Dr.E.Laxmi Lydia
5.	Hadoop Map Reduce Cookbook, SrinathPerera, ThilinaGunarathne
e-Resources	
1.	Hadoop: https://hadoop.apache.org/
2.	Hive: https://cwiki.apache.org/confluence/display/Hive/Home/
3.	Piglatin: https://pig.apache.org/docs/r0.7.0/tutorial.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CT4112	SEC	--	1	2	2	30	70	3 Hrs.
PROMPT ENGINEERING								
(For CSIT)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	Apply iterative prompting for clarity and context.							
2.	Create varied prompts to steer model outputs.							
3.	Construct chain-of-thought and structured prompts.							
4.	Develop retrieval-augmented pipelines to ground outputs.							
5.	Evaluate LLM agents and multimodal apps for ethics and robustness.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1.	Apply prompting techniques for classification, question answering, and code generation.							K3
2.	Design prompts for text generation, debugging, SQL queries, chatbots.							K3
3.	Construct advanced prompts for structured outputs and real-world problem solving.							K3
SYLLABUS								
UNIT-I (10Hrs)	Foundations of Prompt Engineering: Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation							
	Lab Experiments:							
	1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple “Hello, world” prompt to verify model access.							
	2. Baseline vs. Enhanced Prompts: Execute a naïve prompt (“Write a one-paragraph bio of Ada Lovelace.”) and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style.							
	3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: a. Minimal instruction. b. Addition of length and style constraints c. Specification of key content elements (setting and theme)							
	Document how each iteration changes and improves the result.							
	4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt;							

	analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.
UNIT-II (10 Hrs)	Advanced Prompt Patterns & Techniques: Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization. Lab Experiments: 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 3. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., “You are a financial advisor...”); then create a negative prompt to suppress undesired content (e.g., “Do not mention any brand names”); evaluate how each influences the model’s response. 4. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.
UNIT-III (10 Hrs)	Unit III: Structured Output & Reasoning Techniques: Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks Lab Experiments: 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., “List three benefits of daily exercise in a Markdown table with columns ‘Benefit’ and ‘Description.’”); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. 3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., “Let’s think step by step. Explain your reasoning before the final answer.”); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline.

UNIT-IV (10 Hrs)	Unit IV: Retrieval-Augmented Generation & LangChain Workflows: Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval-generation pipeline with an LCEL chain Lab Experiments: 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., “Summarize this text: ...”), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer Test with sample queries and compare factual accuracy against a prompt without retrieval.
UNIT-V (10 Hrs)	Agents, Multimodal AI & Ethical Evaluation: Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to “LLM-as-Judge” for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning Lab Experiments: 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. b. Use an “LLM-as-Judge” prompt (e.g., “Rate these outputs on a scale of 1–5 for clarity

	and correctness.”) to automate evaluation. c. Design a prompt- injection test (e.g., “Ignore previous instructions...”), observe the response, then refine system prompts to mitigate the vulnerability.
Text Books:	
1.	Prompt Engineering for Generative AI, James Phoenix, Mike Taylor, 1st Edition, O’Reilly Media
2.	Generative AI on AWS, Chris Fregly, Antje Barth, 1st Editio, O’Reilly Media
Reference Books:	
1.	Brown, T. B., et al. “ <i>Language Models are Few-Shot Learners.</i> ” Advances in Neural Information Processing Systems (NeurIPS), 2020.
2.	Russell, S., & Norvig, P. “ <i>Artificial Intelligence: A Modern Approach.</i> ” 4th Edition, Pearson, 2020.
e-Resources	
1.	Preview: Secure Computation: Part I SWAYAM
2.	LangChain overview - Docs by LangChain
3.	Prompt engineering OpenAI API



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30		--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	To understanding of the fundamental principles, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, Governments.							
3.	To Develop awareness of constitutional bodies and welfare commissions.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Understand the concepts of the Indian Constitution.							K2
2.	Explain the structure and functions of the Union Government.							K2
3.	Explain the structure, functions of State Government and administrative roles.							K2
4.	Examine local administration and the roles of district, municipal, and Panchayati							K3
5.	Demonstrate the roles of the Election Commission and welfare commissions in governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							
UNIT-III (6Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.							
UNIT-IV (6Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Panchayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass							

	root democracy.
UNIT-V (6Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)

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Regulation: R23	IV / IV - B.Tech. II - Semester								
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CT4201	Project and Internship	PR	0	0	24	12	60	140	200
TOTAL			-	-	24	12	60	140	200



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	I.M.	E.M.	Exam
B23CT4201	PR	--	--	24	12	60	140	3H
PROJECT WORK								
Pre-requisites:: SE and Technical Skills in different Technologies, skills, planning, documentation								
Course Objectives: The main objectives of the course are to								
1.	To provide an opportunity to work in group on a topic / problem / experimentation							
2.	To encourage creative thinking process							
3.	To provide an opportunity to analyze and discuss the results to draw conclusions							
	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Identify a current problem through literature/field/case studies							K3
2.	Identify the objectives and methodology for solving the problem							K3
3.	Design and Develop technology/process for solving the problem							K4
4.	Evaluate the technology/process							K5
SYLLABUS								
	*The object of Project Work is to enable the student to take up investigative study in the broad field of Computer Science and Information Technology, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work. The assignment to normally include: a) Survey and study of published literature on the assigned topic. b) Working out a preliminary approach to the problem relating to the assigned topic. c) Conducting preliminary Analysis/Modelling/Simulation/ Experiment/ Design / Feasibility. d) Preparing a written report on the study conducted for presentation to the department. e) Final Seminar, as oral Presentation before a departmental committee							